Lab 08

Two-Dimensional Arrays in C++

Objective

The objective of this lab is to understand and implement two-dimensional arrays in C++, including their declaration, initialization, and traversal using loops.

Introduction

A two-dimensional array is a collection of elements arranged in rows and columns, similar to a table or matrix. It allows storing multiple values in a structured format and is commonly used in applications such as matrices, grids, and tables.

Theory

A two-dimensional array in C++ is declared as:

```
data_type array_name[rows][columns];
```

Example

```
int matrix[2][3] = {
    {1, 2, 3},
    {4, 5, 6}
};
```

Key Points

- It consists of rows and columns
- Indexing starts from 0
- Access element using:

  array[row][column]
- Nested loops are used to traverse the array

Algorithm:

1. Start
2. Declare a 2D array
3. Input values (optional)
4. Use nested loops to access elements
5. Display the array
6. End

Program:

```
#include <iostream>   // input aur output ke liye
using namespace std;

int main() {

    int rows = 2, cols = 3;   // rows aur columns ki size define ki
    int matrix[2][3];         // 2x3 matrix banayi

    // user se matrix ke elements lena
    cout << "Enter elements of 2x3 matrix:\n";
    for(int i = 0; i < rows; i++) {          // rows ke liye loop
        for(int j = 0; j < cols; j++) {      // columns ke liye loop
            cin >> matrix[i][j];             // user se value input lena
        }
    }

    // matrix display karna
    cout << "\nMatrix is:\n";
    for(int i = 0; i < rows; i++) {          // rows ko print karne ke liye loop
        for(int j = 0; j < cols; j++) {      // columns ko print karne ke liye loop
            cout << matrix[i][j] << "\t";  // har element ko print karna
        }
        cout << endl;}  // next line par jana (new row)

    return 0; // program ka end
        }
}
```

Output:

```
Enter elements of 2x3 matrix:
1 2 3
4 5 6
Matrix is:
1    2    3
4    5    6
```

Lab Tasks

Task 01:

Two-Dimensional Array Declaration and Initialization

• Declare a 2D integer array called ”matrix” with dimensions 3x3.

• Initialize the array with values 1, 2, 3, 4, 5, 6, 7, 8, and 9.

• Display the values of the array elements using nested loops.

Code:

```
#include <iostream>   // For input and output
using namespace std;

int main() {
 int matrix[3][3];

// Create a 3x3 matrix (2D array)
// Start value from 1
int value = 1

// Fill the matrix with numbers from 1 to 9
for (int i = 0; i < 3; i++) {        // Loop for rows
for (int j = 0; j < 3; j++) {    // Loop for columns
matrix[i][j] = value;        // Put value in matrix
value++;                     // Increase value
}
}
 // Print the matrix
    cout << "Matrix elements are:\n";
 for (int i = 0; i < 3; i++) {        // Loop through row
for (int j = 0; j < 3; j++) {    // Loop through columns
 cout << matrix[i][j] << " "; // Print each number
}
 cout << endl;                    // Move to next line after each row
    }
 return 0;  // End of program
}
```

Task 02:

Sum of Rows and Columns in a Two-Dimensional Array

• Declare and initialize a 3x3 integer array with random values.

• Calculate and display the sum of each row and each column separately.

• Display the total sum of all the elements in the array.

Code:

```
#include <iostream>
using namespace std;

int main() {
    // 3x3 array ko values ke sath initialize kar rahe hain
    int arr[3][3] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };
    int totalSum = 0;
 // Har row ka sum nikalna
    cout << "Row sums:\n";
    for (int i = 0; i < 3; i++) {
        int rowSum = 0;
        for (int j = 0; j < 3; j++) {
            rowSum += arr[i][j];      // row ke elements ko add kar rahe hain
            totalSum += arr[i][j];    // total sum mein bhi add kar rahe hain
        }
        cout << "Row " << i + 1 << " sum = " << rowSum << endl;
    }
     // Har column ka sum nikalna
    cout << "\nColumn sums:\n";
    for (int j = 0; j < 3; j++) {
        int colSum = 0;
        for (int i = 0; i < 3; i++) {
            colSum += arr[i][j];   // column ke elements ko add kar rahe hain
        }
        cout << "Column " << j + 1 << " sum = " << colSum << endl;
    }
     // Total sum print karna
    cout << "\nTotal sum of all elements = " << totalSum << endl;
    return 0;}
```

Task 03:

Finding the Maximum Value in a Two-Dimensional Array

- Declare and initialize a 4x4 integer array with random values
- Find and display the maximum value in the array.

Code:

```cpp
#include <iostream>
using namespace std;

int main() {
    // 4x4 array ko values ke sath initialize kar rahe hain
    int arr[4][4] = {
        {12, 5, 7, 9},
        {3, 15, 8, 6},
        {10, 2, 14, 11},
        {4, 18, 1, 13}
    };

    // Pehle element ko maximum maan rahe hain
    int maxValue = arr[0][0];

    // Maximum value dhoondhna
    for (int i = 0; i < 4; i++) {
        for (int j = 0; j < 4; j++) {
            // Agar current element maxValue se bara ho
            if (arr[i][j] > maxValue) {
                maxValue = arr[i][j]; // to maxValue update kar do
            }
        }
    }

    // Maximum value print karna
    cout << "Maximum value in the array = " << maxValue << endl;

    return 0;
}
```

Conclusion:

In this lab, we learned how to declare, initialize, and manipulate two-dimensional arrays in C++. We also practiced using nested loops to input and display matrix elements. This concept is essential for handling structured data like tables and matrices in programming.